Tutorial 7C Solutions

1. Find a **vector equation** of the plane, in **parametric form** and in **scalar product form**, where the plane contains

(a) the points A, B and C with the position vectors $\mathbf{i}+\mathbf{j}-2\mathbf{k}$, $\mathbf{i}+\mathbf{k}$ and $-2\mathbf{i}+\mathbf{j}-3\mathbf{k}$ respectively,

(b) the origin and the line with equation $\mathbf{r}=\begin{pmatrix}2\\0\\-1\end{pmatrix}+\lambda\begin{pmatrix}3\\4\\-1\end{pmatrix}$ where $\lambda\in\mathbb{R}$,

(c) the lines with equations $\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+s\begin{pmatrix}1\\2\\-1\end{pmatrix}$ where $s\in\mathbb{R}$ and $\mathbf{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+t\begin{pmatrix}-1\\1\\-2\end{pmatrix}$ where $t\in\mathbb{R}$,

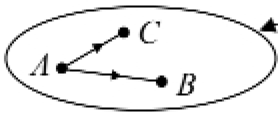
(d) the point $(1,1,1)$ and is parallel to the vectors $\mathbf{i}+2\mathbf{j}+\mathbf{k}$ and $-\mathbf{i}+\mathbf{k}$,

(e) the point $(2,-1,-1)$ and a normal vector $\mathbf{i}-\mathbf{j}+\mathbf{k}$,

(f) the line with equation $\mathbf{r}=\begin{pmatrix}1\\1\\1\end{pmatrix}+\lambda\begin{pmatrix}-1\\2\\-1\end{pmatrix}$ where $\lambda\in\mathbb{R}$ and is perpendicular to the plane: $2x+3y+4z=5$.

Solution

(a) $\overrightarrow{AB}=\overrightarrow{OB}-\overrightarrow{OA}=\begin{pmatrix}1\\0\\1\end{pmatrix}-\begin{pmatrix}1\\1\\-2\end{pmatrix}=\begin{pmatrix}0\\-1\\3\end{pmatrix}$



Always include a sketch to visualise information given.

$\overrightarrow{AC}=\overrightarrow{OC}-\overrightarrow{OA}=\begin{pmatrix}-2\\1\\-3\end{pmatrix}-\begin{pmatrix}1\\1\\-2\end{pmatrix}=\begin{pmatrix}-3\\0\\-1\end{pmatrix}=-\begin{pmatrix}3\\0\\1\end{pmatrix}$

If $\overrightarrow{AC}$ is needed for its direction only, use simplest form $\begin{pmatrix}3\\0\\1\end{pmatrix}$. If actual vector $\overrightarrow{AC}$ is needed, then must use $\begin{pmatrix}-3\\0\\-1\end{pmatrix}$.

Hence a vector equation of plane in parametric form is

$\underset{\sim}{r}=\begin{pmatrix}1\\1\\-2\end{pmatrix}+\lambda\begin{pmatrix}0\\-1\\3\end{pmatrix}+\mu\begin{pmatrix}3\\0\\1\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}.$

Director vectors chosen must be non-parallel.

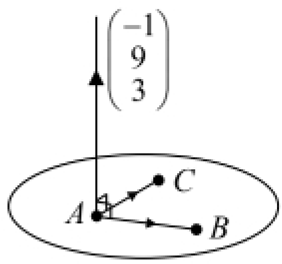
Can also use $\begin{pmatrix}1\\0\\1\end{pmatrix}$ or $\begin{pmatrix}-2\\1\\-3\end{pmatrix}$ as they are position vectors of points that are also on plane.

Thus, vector equation of a plane in parametric form is not unique.

A normal to plane is $\begin{pmatrix}0\\-1\\3\end{pmatrix}\times\begin{pmatrix}3\\0\\1\end{pmatrix}=\begin{pmatrix}-1\\9\\3\end{pmatrix}$

$\therefore\ \underset{\sim}{r}\cdot\begin{pmatrix}-1\\9\\3\end{pmatrix}=\begin{pmatrix}1\\1\\-2\end{pmatrix}\cdot\begin{pmatrix}-1\\9\\3\end{pmatrix}=-1+9-6=2$

Using $\underset{\sim}{r}\cdot\underset{\sim}{n}=\underset{\sim}{a}\cdot\underset{\sim}{n}$



Hence equation of plane in scalar product form is $\underset{\sim}{r}\bullet\begin{pmatrix}-1\\9\\3\end{pmatrix}=2$.

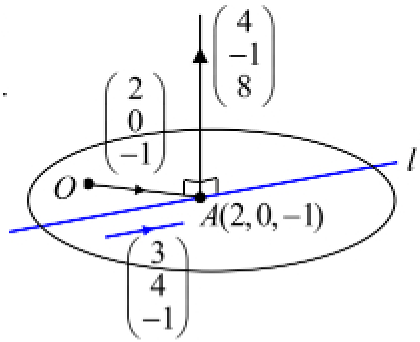
(b) Let O and A represents origin and point $(2,0,-1)$ respectively.

Since both O and A are on plane, $\overrightarrow{OA}=\begin{pmatrix}2\\0\\-1\end{pmatrix}$ is on plane and thus parallel to plane.

Hence a vector equation of plane in parametric form is

$$\underset{\sim}{r}=\begin{pmatrix}0\\0\\0\end{pmatrix}+\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}3\\4\\-1\end{pmatrix}=\lambda\begin{pmatrix}2\\0\\-1\end{pmatrix}+\mu\begin{pmatrix}3\\4\\-1\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}.$$

Origin O is on plane.



A normal to plane is $\begin{pmatrix}2\\0\\-1\end{pmatrix}\times\begin{pmatrix}3\\4\\-1\end{pmatrix}=\begin{pmatrix}4\\-1\\8\end{pmatrix}$

$$\therefore\ \underset{\sim}{r}\bullet\begin{pmatrix}4\\-1\\8\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}\bullet\begin{pmatrix}4\\-1\\8\end{pmatrix}=0$$

Using $\underset{\sim}{r}\bullet\underset{\sim}{n}=\underset{\sim}{a}\bullet\underset{\sim}{n}$

Hence equation of plane in scalar product form is $\underset{\sim}{r}.\begin{pmatrix}4\\-1\\8\end{pmatrix}=0$.

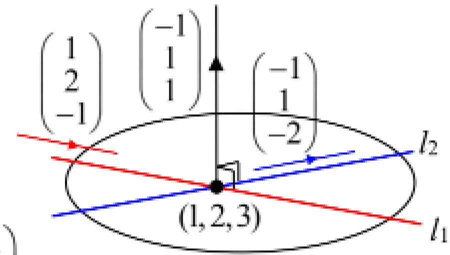
(c) Let $l_1:\ \underset{\sim}{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+s\begin{pmatrix}1\\2\\-1\end{pmatrix}$, $l_2:\ \underset{\sim}{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+t\begin{pmatrix}-1\\1\\-2\end{pmatrix}$, $s,t\in\mathbb{R}$.

Note that $\begin{pmatrix}1\\1\\0\end{pmatrix}$ is common to both l_1 and l_2 and thus $(1,1,0)$ is point of intersection between l_1 and l_2 .

A vector equation of plane in parametric form is

$$\underset{\sim}{r}=\begin{pmatrix}1\\2\\3\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\-1\end{pmatrix}+\mu\begin{pmatrix}-1\\1\\-2\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}.$$

A normal to plane is $\begin{pmatrix}1\\2\\-1\end{pmatrix}\times\begin{pmatrix}-1\\1\\-2\end{pmatrix}=\begin{pmatrix}-3\\3\\3\end{pmatrix}=3\begin{pmatrix}-1\\1\\1\end{pmatrix}$



$$\therefore\ \underset{\sim}{r}\bullet\begin{pmatrix}-1\\1\\1\end{pmatrix}=\begin{pmatrix}1\\2\\3\end{pmatrix}\bullet\begin{pmatrix}-1\\1\\1\end{pmatrix}=-1+2+3=4$$

Using $\underset{\sim}{r}\bullet\underset{\sim}{n}=\underset{\sim}{a}\bullet\underset{\sim}{n}$

Hence equation of plane in scalar product form is $\underset{\sim}{r}\bullet\begin{pmatrix}-1\\1\\1\end{pmatrix}=4$.

(d) A vector equation of plane in parametric form is $\underset{\sim}{r}=\begin{pmatrix}1\\1\\1\end{pmatrix}+\lambda\begin{pmatrix}1\\2\\1\end{pmatrix}+\mu\begin{pmatrix}-1\\0\\1\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}$.

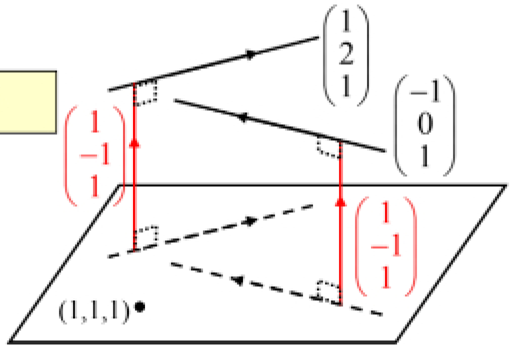
$(0,1,1)$ lies on plane

A normal to plane is $\begin{pmatrix}1\\2\\1\end{pmatrix}\times\begin{pmatrix}-1\\0\\1\end{pmatrix}=\begin{pmatrix}2\\-2\\2\end{pmatrix}=2\begin{pmatrix}1\\-1\\1\end{pmatrix}$

$\therefore\ \underset{\sim}{r}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=\begin{pmatrix}1\\1\\1\end{pmatrix}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=1-1+1=1$

Using $\underset{\sim}{r}\bullet\underset{\sim}{n}=\underset{\sim}{a}\bullet\underset{\sim}{n}$

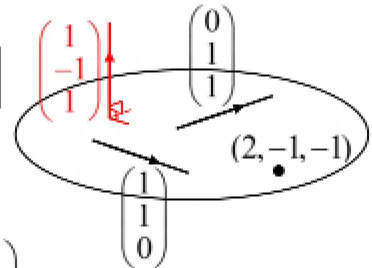
Hence equation of plane in scalar product form is $\underset{\sim}{r}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=1$



(e) Using $\underset{\sim}{r}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=\begin{pmatrix}2\\-1\\-1\end{pmatrix}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=2+1-1=2$

Using $\underset{\sim}{r}\bullet\underset{\sim}{n}=\underset{\sim}{a}\bullet\underset{\sim}{n}$

Hence equation of plane in scalar product form is $\underset{\sim}{r}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=2$



By inspection, $\begin{pmatrix}1\\1\\0\end{pmatrix}$ and $\begin{pmatrix}0\\1\\1\end{pmatrix}$ are perpendicular to $\begin{pmatrix}1\\-1\\1\end{pmatrix}$.

Hence an equation of plane in parametric form is

$\underset{\sim}{r}=\begin{pmatrix}2\\-1\\-1\end{pmatrix}+\lambda\begin{pmatrix}1\\1\\0\end{pmatrix}+\mu\begin{pmatrix}0\\1\\1\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}$.

If a vector $\underset{\sim}{r}$ is parallel to the plane, then $\underset{\sim}{r}$ is perpendicular to $\begin{pmatrix}1\\-1\\1\end{pmatrix}$, and $\underset{\sim}{r}.\begin{pmatrix}1\\-1\\1\end{pmatrix}=0$.
We can then obtain suitable $\underset{\sim}{r}$ by inspection or trial and error.
Here, $\begin{pmatrix}1\\1\\0\end{pmatrix}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=0$ and $\begin{pmatrix}0\\1\\1\end{pmatrix}\bullet\begin{pmatrix}1\\-1\\1\end{pmatrix}=0$.
Note that the 2 direction vectors in parametric form must be non-parallel.

(f) Equation of given plane is
$2x+3y+4z=5$
$\therefore$ equation of given plane in scalar product form is $\underset{\sim}{r}\bullet\begin{pmatrix}2\\3\\4\end{pmatrix}=5$

Vector equation of required plane in parametric form is

$\underset{\sim}{r}=\begin{pmatrix}1\\1\\1\end{pmatrix}+\lambda\begin{pmatrix}-1\\2\\-1\end{pmatrix}+\mu\begin{pmatrix}2\\3\\4\end{pmatrix},\quad \lambda,\mu\in\mathbb{R}$.

Note that $(1,1,1)$ lies on required plane.

Both are parallel to required plane, so are suitable as direction vectors in parametric form.